

Full Stack Case Study

Chatify App

Designed By
Houssni Msehel

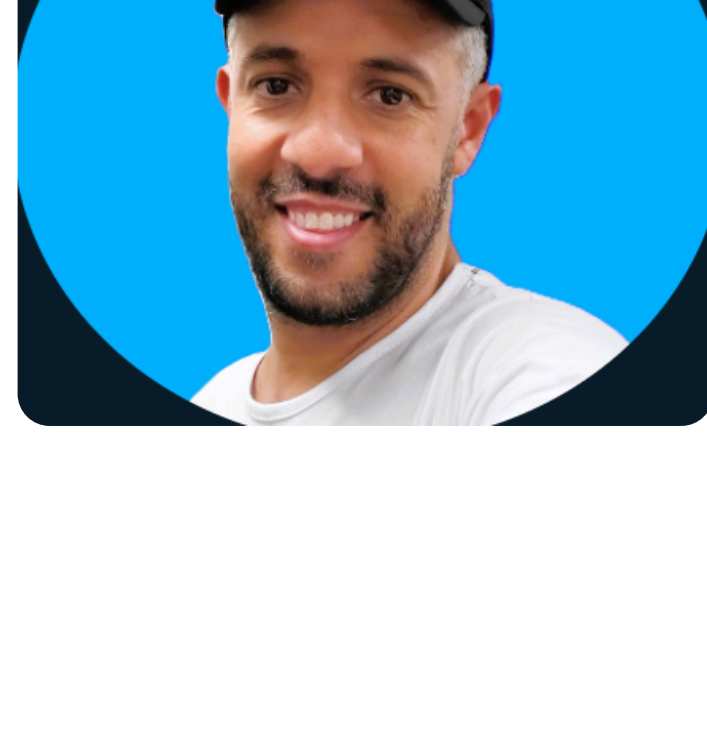
Chatify

Houssni

Choose Background Color

Start Chatting

Github
Link



Hi, I'm **Houssni**! I like making things accessible for the everyday user. I specialize in developing apps for tech companies and retail brands. In my free time, I run creative workshops for kids as a way to give back to the community.

Chatify

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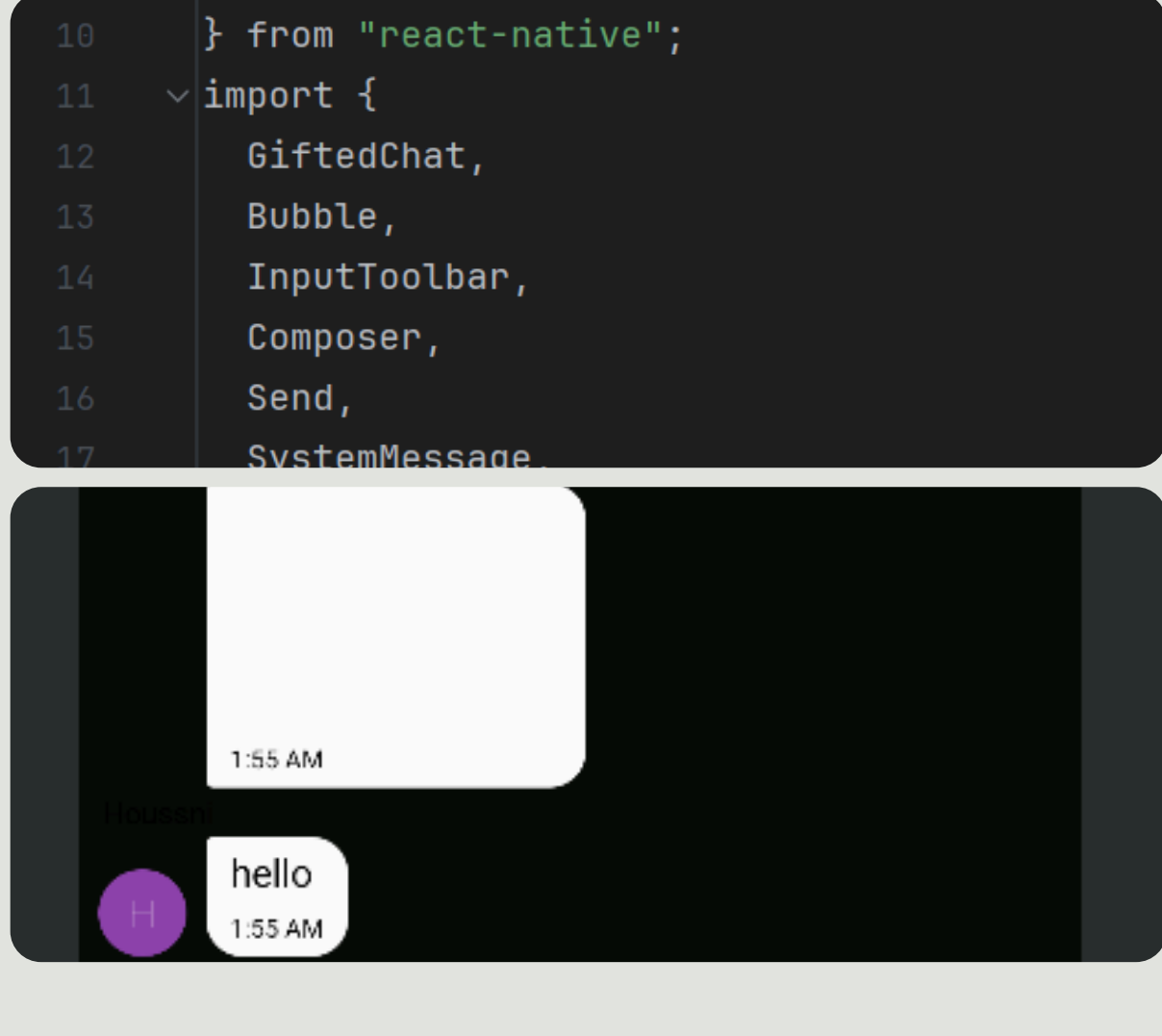
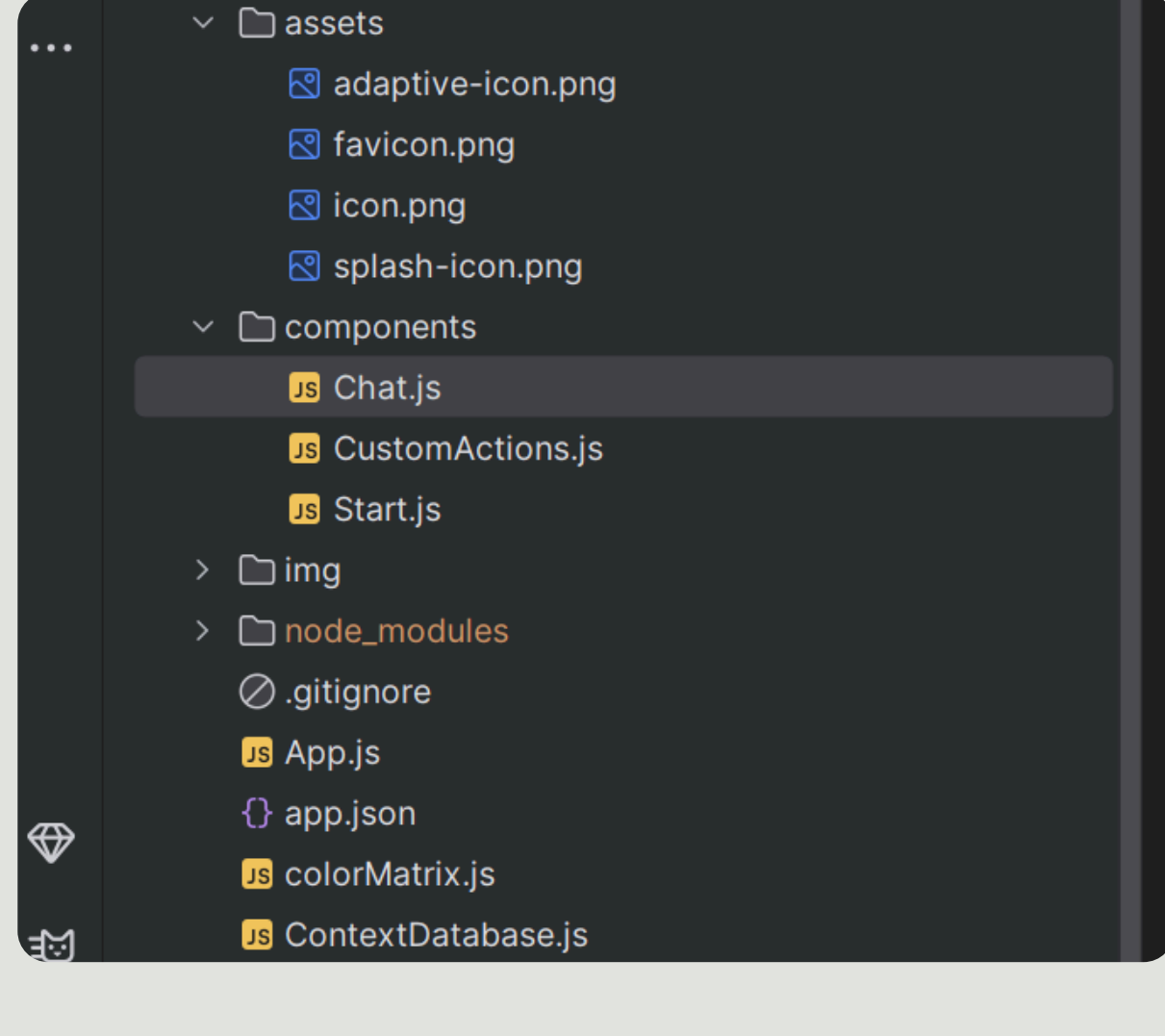
Project Overview

Chatify is a native Android messaging application designed to provide users with a seamless and personalized chat experience. The project involved setting up a robust development environment and building a functional chat interface with customizable themes.

- Project Type: Native Android Mobile Application
- My Role: Android Developer (Full Project Lifecycle)
- Tools & Technologies: Android Studio, Android SDK, Kotlin, Android Emulator

Project gallery

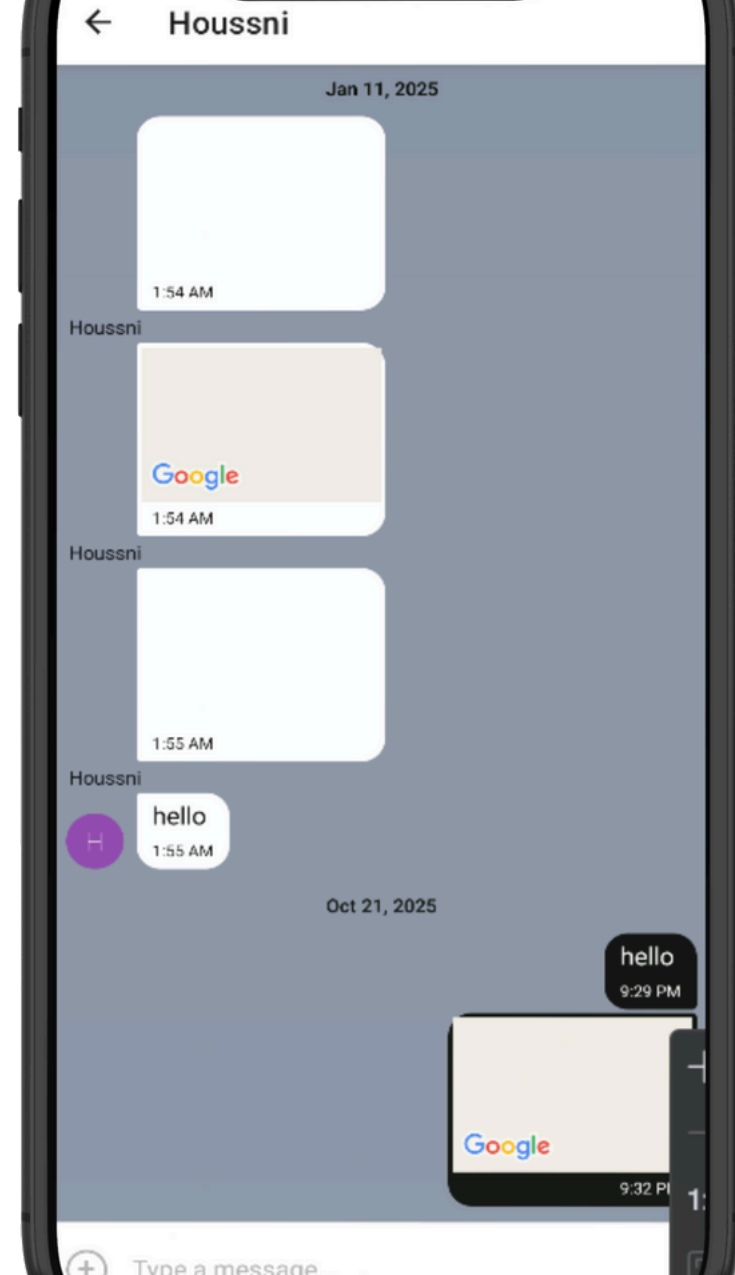
"My designs are all about user-friendliness and accessibility."



Problem Statement

Modern users expect messaging apps that are not only functional but also visually appealing and personalized. Many existing solutions either lack customization options or have complex interfaces. Chatify was developed to address these needs by providing:

- A clean, intuitive chatting interface
- Personalization through background color customization
- A smooth user experience on various Android devices



Solution & Implementation

Phase1

Development Environment Setup

The foundation of any successful Android project begins with proper environment configuration. As shown in the screenshots:

Android SDK Configuration:

Configured Android SDK location at

C:\Users\mingit\AppData\Local\Android\Sdk

Installed necessary SDK components including:

Android API Level 32 platforms and sources

Google APIs ARM 64 v8a System Image

Android SDK Platform Tools

Set up Android Emulator with system images for comprehensive testing

Key Technical Decisions:

Selected API Level 32 for optimal compatibility and feature support

Configured multiple system images (ARM, x86_64) for cross-device testing

Implemented Google APIs for extended functionality

Solution & Implementation

Phase 2

Core Features Implemented:

Clean Chat Interface: Minimalist design focusing on message readability

Background Customization: Users can personalize chat backgrounds

Real-time Messaging: Basic chat functionality implementation

Device Compatibility: Optimized for various screen sizes and architectures

User Interface Components:

Main chat activity with message bubbles

Background color selector interface

Responsive layout adapting to different device sizes

Technical Challenges & Solutions

Challenge 1: Environment Configuration Complexity

Problem: Setting up the Android SDK with correct system images and dependencies can be error-prone, especially with different processor architectures (ARM vs x86_64).

Solution:

Meticulously configured SDK platforms and tools through Android Studio's SDK Manager

Installed multiple system images to ensure comprehensive testing coverage

Verified installation through detailed package parsing and validation

Challenge 2: Emulator Performance Optimization

Problem: Android Emulator hypervisor driver installation encountered errors (Error 4294967201), potentially impacting testing efficiency.

Solution:

Troubleshooting hypervisor conflicts

Utilizing multiple virtual device configurations

Ensuring successful creation of "Medium_Phone_API_35" test device

Conculusion

Chatify demonstrates the complete lifecycle of Android application development—from environment setup to functional implementation. The project highlights the importance of proper SDK configuration, emulator management, and user-centric design in creating engaging mobile experiences. The successful creation of a customizable chat interface lays the foundation for future enhancements and more complex feature implementations.

Let's work together!

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